問題 171

次の関数について，偶関数か奇関数かを調べよ．

解答:

$f(-x)=f(x)$ → 偶（y 軸対称），$f(-x)=-f(x)$ → 奇（原点対称）．

(1) $3x^2$ → 偶　(2) $x-1$ → どちらでもない　(3) $-2x$ → 奇

(4) x^2+x → どちらでもない　(5) x^4+5x^2-3 → 偶

(6) $(x-1)^5$ → どちらでもない　(7) $\dfrac{1}{4}x^3$ → 奇

(8) $\dfrac{1}{x^2}$ → 偶　(9) $\dfrac{(x^2+3)^2}{x}$ → 奇

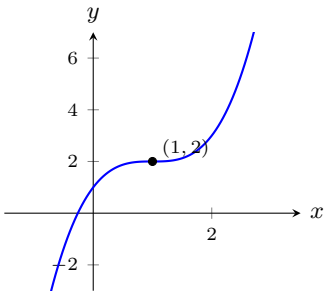
問題 172

次の関数のグラフをかけ．

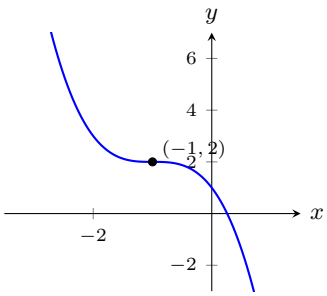
解答:

基本関数 $y=x^n$ の平行移動・反転で描く．

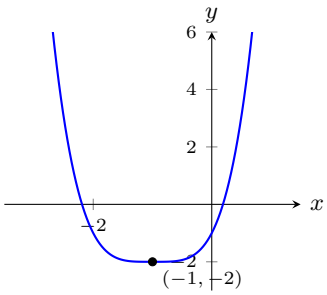
(1) $y=(x-1)^3+2$：変曲点 $(1,2)$



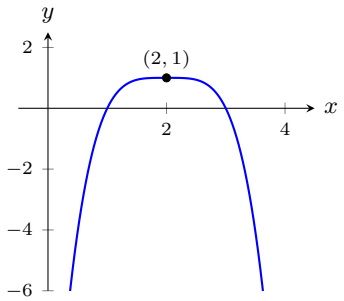
(2) $y=-(x+1)^3+2$：変曲点 $(-1,2)$



(3) $y=(x+1)^4-2$：最小点 $(-1,-2)$



(4) $y=-(x-2)^4+1$：最大点 $(2,1)$



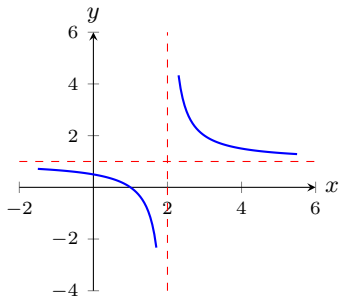
問題 173

次の関数の定義域と値域を求め，グラフをかけ．また，漸近線を求めよ．

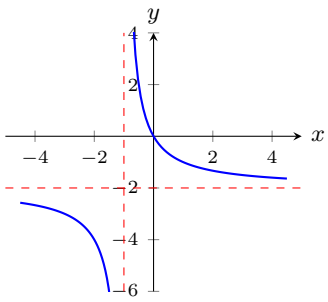
解答:

$y=\dfrac{k}{x-p}+q$ → 漸近線 $x=p$，$y=q$.

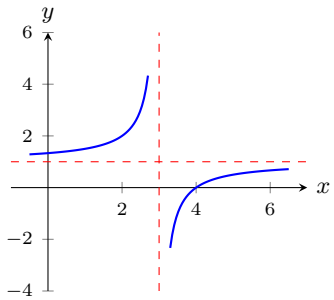
(1) $y=\dfrac{1}{x-2}+1$　漸近線：$x=2$，$y=1$



(2) $y=\dfrac{2}{x+1}-2$　漸近線：$x=-1$，$y=-2$



(3) $y=-\dfrac{1}{x-3}+1$　漸近線：$x=3$，$y=1$



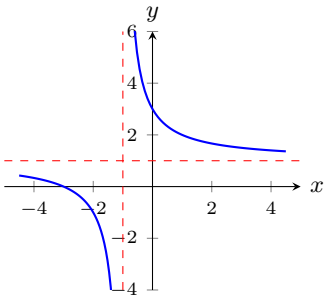
問題 174

次の関数のグラフをかけ．また，定義域と値域および漸近線を求めよ．

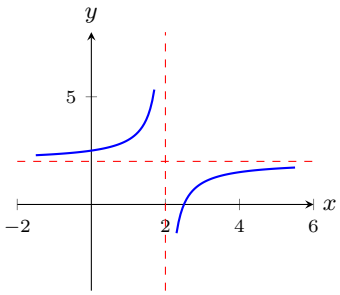
解答:

分子 ÷ 分母で帯分数に変形する．

(1) $y = \dfrac{x+3}{x+1} = 1 + \dfrac{2}{x+1}$　漸近線：$x = -1$，$y = 1$



(2) $y = \dfrac{2x-5}{x-2} = 2 - \dfrac{1}{x-2}$　漸近線：$x = 2$，$y = 2$



問題 175

次の関数の定義域を求めよ．

解答:

(1) $\sqrt{x-2}$：$x \geqq 2$　(2) $\dfrac{-3}{\sqrt{x+1}}$：$x > -1$　(3) $\sqrt{-x^2+x+6}$：$-2 \leqq x \leqq 3$

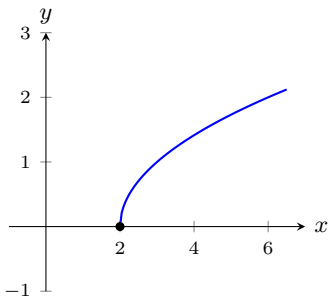
問題 176

次の関数の定義域と値域を求め，グラフをかけ．

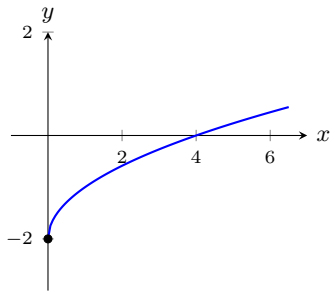
解答:

$y = \sqrt{x}$ の平行移動で考える．

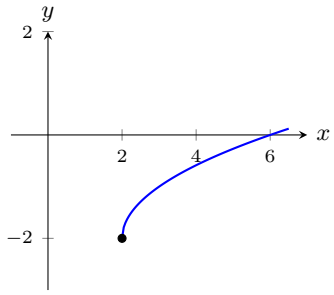
(1) $y = \sqrt{x-2}$：始点 $(2, 0)$．定義域 $x \geqq 2$，値域 $y \geqq 0$



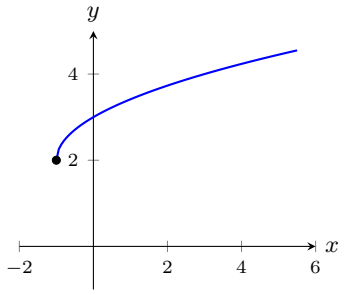
(2) $y = \sqrt{x} - 2$：始点 $(0, -2)$．定義域 $x \geqq 0$，値域 $y \geqq -2$



(3) $y = \sqrt{x-2} - 2$：始点 $(2, -2)$．定義域 $x \geqq 2$，値域 $y \geqq -2$



(4) $y = \sqrt{x+1} + 2$：始点 $(-1, 2)$．定義域 $x \geqq -1$，値域 $y \geqq 2$



問題 177

次の直線または点に関して，$y = x^3 + 3x^2$ と対称なグラフをもつ関数を求めよ．

解答:

(1) x 軸：$y = -x^3 - 3x^2$

(2) y 軸：$y = -x^3 + 3x^2$

(3) 原点：$y = x^3 - 3x^2$

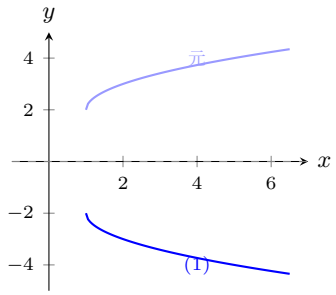
問題 178

関数 $y = \sqrt{x-1} + 2$ のグラフを用いて，次の関数のグラフをかけ．

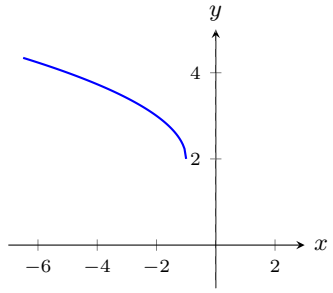
解答:

元のグラフとの対称関係を読み取る．

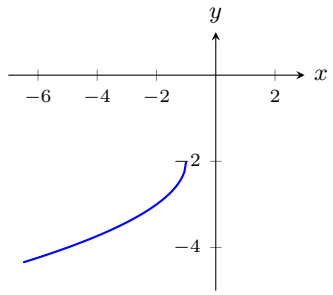
(1) $y = -\sqrt{x-1} - 2$：x 軸対称



(2) $y=\sqrt{-x-1}+2$：y 軸対称



(3) $y=-\sqrt{-x-1}-2$：原点対称



問題 179

次の関数の逆関数を求め，その逆関数の定義域と値域を求めよ．

解答:

$y=f(x)$ を x について解き，x と y を入れ替える．

(1) $y=\dfrac{1}{x-2}\to y=\dfrac{1}{x}+2$　定義域 $x\neq 0$，値域 $y\neq 2$

(2) $y=x^2-3\ (x\geqq 0)\to y=\sqrt{x+3}$　定義域 $x\geqq -3$，値域 $y\geqq 0$

(3) $y=-\sqrt{x}\to y=x^2\ (x\leqq 0)$　定義域 $x\leqq 0$，値域 $y\geqq 0$

(4) $y=\sqrt{2-x}-1\to y=-x^2-2x+1\ (x\geqq -1)$

定義域 $x\geqq -1$，値域 $y\leqq 2$